Safety Engineering

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A car's safety has three jobs: help the driver **avoid** a crash, help everyone **survive** one, and help rescuers afterwards. Avoidance comes from brakes, tyres, ABS and stability control. Survival comes from a rigid passenger cell, crumple zones that absorb energy, seatbelts and airbags. In India, crash ratings from Bharat NCAP (government, since 2023) and Global NCAP (independent) now let buyers compare cars on tested performance instead of marketing. A star rating is meaningful only for the exact version tested and the protocol year it was tested under.

1. The physics: why speed is the enemy

In a crash, the car's kinetic energy (½ × mass × speed²) has to go somewhere. It goes into bending metal, which is the safety engineer's job to arrange. Doubling the speed quadruples the energy. Bharat NCAP's frontal offset test runs at 64 km/h; a crash at 120 km/h carries about 3.5 times that energy.* No structure can absorb that safely, which is why speed limits matter more than any feature.

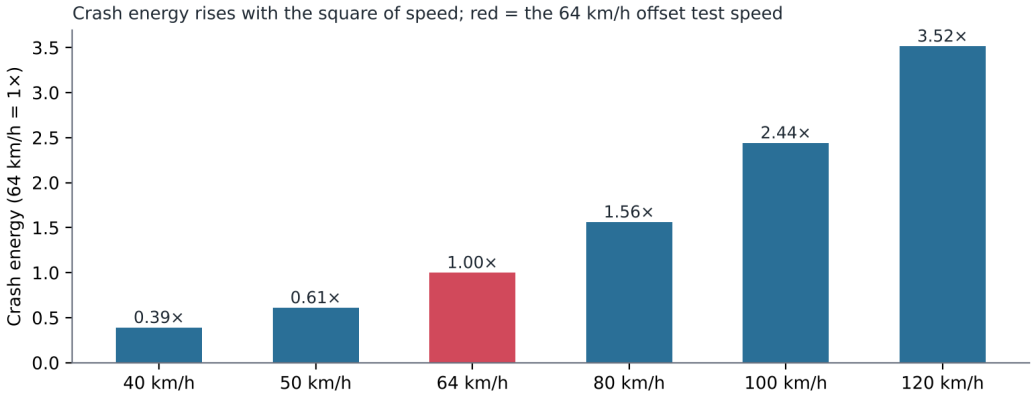


Fig 1. Relative crash energy against the 64 km/h test speed. Energy = ½ mv², so it scales with speed squared.

2. Three layers of safety

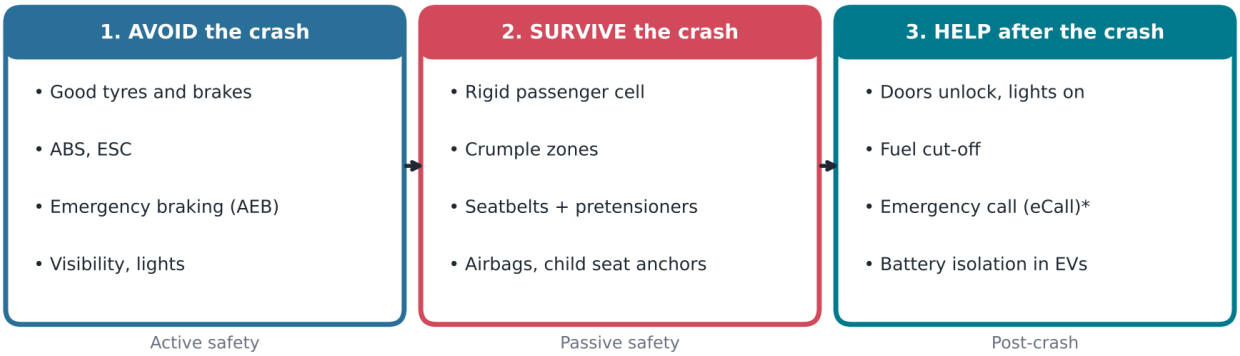


Fig 2. Safety as three layers: avoid, survive, help after the crash. (*eCall is not yet widely fitted in India.)

3. How a car survives a crash

- **Rigid passenger cell:** the cabin is built from very strong steel (often high-strength or hot-stamped grades) so it does not collapse onto occupants.
- **Crumple zones:** the front and rear are designed to fold progressively, absorbing energy over a longer time. Stopping over a longer time means lower force on the body.
- **Seatbelts with pretensioners and load limiters:** the belt tightens in the first milliseconds and then lets out a little so the chest is not injured. The belt is the single most important restraint; airbags assume you are wearing it.
- **Airbags:** a front airbag cushions the head and chest; side and curtain airbags protect in side impacts and rollovers. India's rules for six airbags have been proposed and deferred more than once; many manufacturers now fit six as standard on popular models.*
- **Child restraints (ISOFIX):** standard anchor points let a child seat clip rigidly into the car. Child occupant protection is scored separately in NCAP tests.

4. How a car avoids a crash

- **ABS** prevents wheel lock under hard braking; **ESC** stops skids by braking single wheels. Global NCAP has pushed ESC as a key feature, and it is now rewarded in the protocol (see below).
- **Autonomous emergency braking (AEB)** uses a radar or camera to brake automatically when a collision is near. It is part of the driver-assistance systems in Part 8.
- **Tyres and brakes in good condition** matter more than any electronic feature; Part 4 covers stopping distances.

5. Crash tests: how to read the stars

Global NCAP, a UK-based charity, started crash-testing popular Indian cars in 2014 under its Safer Cars for India campaign. Its first round tested five cars, including the Swift, with no frontal airbags fitted; all scored zero stars for adult protection. Results pushed Toyota and Volkswagen to make frontal airbags standard across their Indian lineups.

Bharat NCAP is India's own programme. It was launched by the Ministry of Road Transport and Highways on 22 August 2023, with operations starting on 1 October 2023 and official testing beginning in December 2023. Tests include a 64 km/h frontal offset crash, side impact, pedestrian protection and child dummy tests. The first 5-star results went to the Tata Harrier and Safari. It is voluntary: makers choose which cars to submit.

Car and test	Protocol / year	Result
Maruti Swift (3rd gen)	Global NCAP 2.0, 2022	Adult 19.19/34, child 16.68/49
Maruti Dzire (4th gen)	Global NCAP, Nov 2024	Adult 31.24/34 (5 stars); first Maruti to score 5 stars; child 39.20/49
Tata Punch (petrol)	Global NCAP, 2021	Adult 16.45/17 (5 stars, older protocol); child 40.89/49
Tata Punch.ev	Bharat NCAP, 2024	Adult 31.46/32 (5 stars); child 45.00/49 (5 stars)
VW Taigun, Skoda Kushaq	Global NCAP, Oct 2022	First cars to reach 5 stars under the updated protocol
Hyundai Creta, i20	Global NCAP, 2022	3 stars each

Why scores aren't directly comparable. Global NCAP changed its protocol in July 2022 (adding side-impact protection and rear seatbelt reminders, and making ESC and side head airbags requirements for higher ratings) and again in August 2025 (rewarding safety equipment only if fitted as standard). A 5-star result in 2021 and one in 2024 reflect different tests. Always check which protocol, which variant (airbag count differs by trim) and which year.

6. What it means for a buyer

- Choose the variant that matches the tested one: the rating usually applies to a specific trim and airbag count.
- Check which protocol and year the rating came from; newer protocols are stricter.
- Prefer cars with six airbags, ESC and ISOFIX as standard, not as paid extras.
- Wear the belt in the rear seat too; most rear passengers in India still do not.

Summary

- Safety has three layers: avoid the crash, survive it, help afterwards.
- Crash energy rises with speed squared, so a structure that is safe at 64 km/h cannot protect at 120 km/h.
- A rigid cell, crumple zones, belts with pretensioners and airbags protect occupants together.
- ABS, ESC and AEB help avoid the crash.
- Bharat NCAP (since 2023) and Global NCAP rate cars; ratings depend on the variant, the protocol and the year.
- Prefer six airbags, ESC and ISOFIX as standard.

*Unverified or general knowledge: the 3.5× comparison is arithmetic ($(120/64)^2$); status of the six-airbag mandate and which models fit six as standard, eCall availability and the seatbelt-usage remark are general knowledge, not checked against official documents. Crash results: Wikipedia entries for Bharat NCAP, Global NCAP, Maruti Dzire, Tata Punch (retrieved November 2026); the Swift row shows scores only because no star rating was confirmed from the sources. Protocol history: Global NCAP (via Wikipedia).

See also: Part 4 (brakes and chassis). Next: Part 6, Electric vehicles.